

1. Workload

2. ISA & ABI

3. Compute Org

4. Memory & NoC

5. Power Intent

6. Compiler/RT

7. Physical Node

8. Verification



CHAPTER STACK SCOPE — FULL STACK ACTIVE (LAYERS 1-8)

Architecture 2.0 Workstream Handoff

1. System Intent

- Lighthouse Product Prompt
- 3 W TDP Mobile XR Target
- RV64GCV + Vector + NPU
- TSMC 3 nm LP Process
- Hard Area/Power Bounds

2. AI Design System

- Foundation Architecture Model
- Multi-Objective Candidate DSE
- Automated Code/RTL Gen
- Diagnostic Triage Engine
- Versioned Project Record

3. Independent Checks

- gem5 / SCALE-Sim Simulators
- Verilator RTL Regression
- OpenROAD Physical Signoff
- STA Slack & Thermal Checks
- Qualified Evidence Card

4. Human Authority

- Senior Architect Review
- Commitment Authority

Verdict Gate:

ADVANCE / REJECT

Telemetry & Failure Diagnostic Feedback